

```
In [1]: import pandas as pd

df = pd.read_csv('usarrests.csv')
df = df.rename(columns={df.columns[0]: 'State'})
print(df.describe())
```

	Murder	Assault	UrbanPop
count	50.00000	49.000000	50.00000
mean	7.78800	182.183673	74.20000
std	4.35551	130.877435	73.40828
min	0.80000	45.000000	6.00000
25%	4.07500	109.000000	53.25000
50%	7.25000	159.000000	66.00000
75%	11.25000	249.000000	77.75000
max	17.40000	879.000000	570.00000

```
In [2]: df.describe()
```

```
Out[2]:
```

	Murder	Assault	UrbanPop
count	50.00000	49.000000	50.00000
mean	7.78800	182.183673	74.20000
std	4.35551	130.877435	73.40828
min	0.80000	45.000000	6.00000
25%	4.07500	109.000000	53.25000
50%	7.25000	159.000000	66.00000
75%	11.25000	249.000000	77.75000
max	17.40000	879.000000	570.00000

Immediate observations

The max UrbanPop(570) and Assault count(49 vs 50 for murder and urbanpop) clearly have issues. The UrbanPop min and Assault max look a little extreme.

```
In [3]: flag_urbanpop_range = df[(df.UrbanPop > 100) | (df.UrbanPop < 0)]
flag_urbanpop_range
```

```
Out[3]:
```

	State	Murder	Assault	UrbanPop
14	Iowa	2.2	56.0	570

flag 1

The UrbanPop value cannot be more than 100 because it is a percentage of the states population. An Urbanpop of 570 is impossible. Action: Correct the value based on the original source or set the value to NaN if that's not possible. It looks like it could have been a decimal typo, so it could have been 57.0.

```
In [4]: df = df.drop(flag_urbanpop_range.index)
df = df.reset_index(drop=True)
```

```
In [5]: df.describe()
```

```
Out[5]:
```

	Murder	Assault	UrbanPop
count	49.000000	48.000000	49.000000
mean	7.902041	184.812500	64.081633
std	4.324566	130.948529	16.592966
min	0.800000	45.000000	6.000000
25%	4.300000	109.000000	53.000000
50%	7.300000	159.000000	66.000000
75%	11.300000	249.750000	77.000000
max	17.400000	879.000000	91.000000

```
In [6]: flag_urbanpop_low = df.sort_values('UrbanPop').head(5)
flag_urbanpop_low
```

```
Out[6]:
```

	State	Murder	Assault	UrbanPop
30	New York	11.1	254.0	6
43	Vermont	2.2	48.0	32
46	West Virginia	5.7	81.0	39
32	North Dakota	0.8	45.0	44
22	Mississippi	16.1	259.0	44

flag 2

New York having an UrbanPop of 6 is extreme and probably not accurate. Logically it is likely incorrect, but it is technically a possible number. Action: Compare it to the original source to see if it is an error, but don't automatically drop it because it could technically be possible.

```
In [7]: flag_missing = df[df.isna().any(axis=1)]
flag_missing
```

Out [7]:

	State	Murder	Assault	UrbanPop
9	Georgia	17.4	NaN	60

flag 3

Georgia is missing a recorded value for assault. This does not seem to be significantly skewing any of the summary table values. Action: Leave it as NaN.

```
In [8]: flag_assault_outlier = df.sort_values('Assault', ascending=False).head(5)
flag_assault_outlier
```

Out [8]:

	State	Murder	Assault	UrbanPop
38	South Carolina	14.4	879.0	48
31	North Carolina	13.0	337.0	45
8	Florida	15.4	335.0	80
18	Maryland	11.3	300.0	67
2	Arizona	8.1	294.0	80

flag 4

South Carolina's assault value is extreme. Assault's mean is ~185 with a std of ~131, so 879 is over 5 standard deviations out and nearly 2.6× the next-highest state. It is very high, but it is possible. Action: Check the original source, but otherwise leave it.